Protein coding gene MYO1E drives significant differences in white matter microstructure between autistic individuals and controls

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Introduction

White matter, composed of myelinated axons, is essential for brain connectivity and communication. Recent studies (Newman et al., 2024; Ressa et al., 2024) have identified that boys and girls with ASD exhibit pronounced differences in white matter microstructure compared to neurotypical children, and these pronounced disruptions in white matter microstructure are linked to cognitive and behavioral difficulties in ASD. However, the identification of various genes linked to these white matter abnormalities has yet to be uncovered. The purpose of this research study is to identify various protein-coding genes that influence white matter microstructural abnormalities in children with ASD, when these individuals with ASD are compared to neurotypical children.

Figure 1: Visualizing White Matter Microstructural Networks between ASD & Controls

Figure 1a: ASD (n = 148)

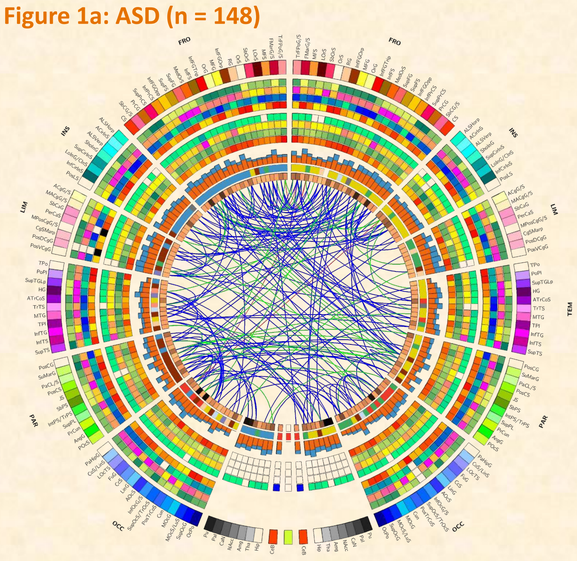
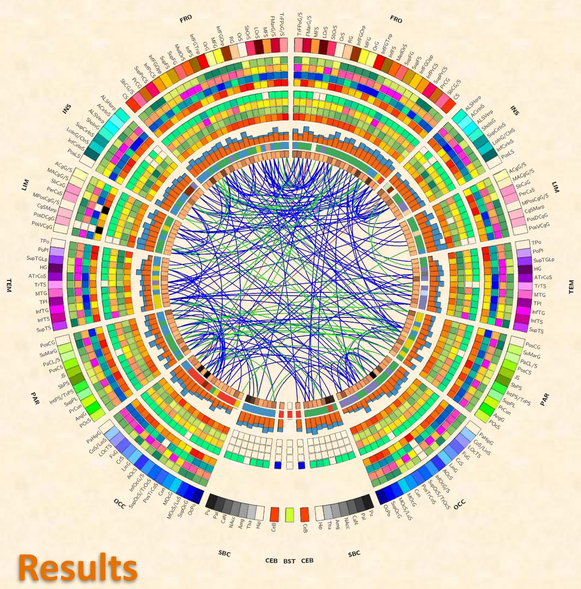


Figure 1b: TD (n = 175)



Average connectograms for each cohort display, from outer to inner rings, various brain atlas reference spaces, mean cortical morphometrics, functional connectivity metrics, regional white matter connectivity degrees (DTI), and interregional connections. Connections are drawn only from regions with statistically significant case-control differences in average white matter microstructural similarity ($p < 0.05$). Figure 1a visualizes ASD individuals, while Figure 1b represents typically developing controls. Line opacity represents connection density, while the color indicates Z--scored wm-MS: blue for ±1.96, green for ±2.5, and red for ±3.25.

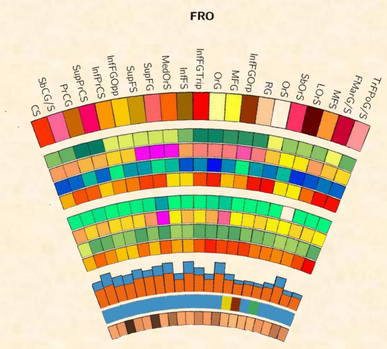


Figure 1c: Visualization Key

1. **Brain Region** *(Outer Ring)*: Brain areas analyzed.
2. **Atlases** *(Rings 2–5)*:
   - **Campbell, Brodmann, Von Economo, Flechsig**: Brain region classifications by each atlas.
3. **FreeSurfer Outputs** *(Rings 6–9)*:
   - **Volume, Area, Thickness, Mean Curvature**: Measurements from FreeSurfer.
4. **Connectivity Metrics** ***(Bars)***:
   - **Resting State Degree, Participation, Modularity, DTI Degree**: Functional and structural connectivity metrics.

Results

Identifying Case-Control Differences in White Matter Microstructure in Various Brain Regions: Results from linear modeling demonstrate that the following brain regions yielded statistically significant p-values after false discovery rate (FDR) corrections, indicating that boys and girls with ASD yield abnormal white matter microstructural features in these regions: inferior occipital gyrus ($p = 0.0226$), middle occipital gyrus ($p = 0.0196$), superior occipital gyrus ($p = 0.0226$), inferior temporal gyrus ($p = 0.0226$), middle temporal gyrus ($p = 0.0469$), posterior lateral fissure ($p = 0.0458$), posterior collateral transverse sulcus ($p = 0.0226$), planum temporale ($p = 0.0303$), superior temporal gyrus ($p = 0.0448$), intraparietal sulcus ($p = 0.0226$), and superior temporal sulcus ($p = 0.0448$).

Identifying Genes Driving Case-Control Differences in White Matter Microstructure: Results from the partial-least squares regression analysis identified the gene MYO1E as a significant factor that drives case-control differences in white matter microstructure between ASD and neurotypical individuals (PLS1 Z-score > 2, PLS2 Z-score > 3) . Furthermore, the protein-coding gene MYO1E, has been considered an important candidate gene associated with ASD risk (SFARI gene score: 2).

Conclusion & Discussion

Case-Control Differences in Key Brain Regions Linked to Cognitive and Behavioral Difficulties in ASD: Our findings reveal significant associations between ASD and multiple cortical regions, with p-values indicating involvement across the inferior, middle, and superior occipital gyri as well as the inferior, middle, and superior temporal gyri ($p < 0.05$). These regions are connected to various white matter tracts, including the inferior fronto-occipital fasciculus (IFOF), inferior longitudinal fasciculus (ILF), and superior longitudinal fasciculus (SLF), which are essential for semantic processing, face and emotion recognition, and visuospatial attention. These tracts have previously been linked to ASD (Andica et al., 2021; Jou et al., 2016).

MYO1E Gene as a Major Driver of Microstructural Differences in ASD: The role of the protein-coding gene, MYO1E, as a significant driver of microstructural differences in ASD is consistent with the fact that it has been scored as a strong candidate gene for ASD, according to the Simons Foundation for Autism Research (Abrahams et al., 2013). Furthermore, according to the Human Protein Atlas (Uhlén et al., 2015), this gene is substantially expressed in white matter (nTPM > 115). Moreover, MYO1E is a myosin protein expressed significantly at different stages of oligodendrocyte (OL) development, and thus has been implicated in oligodendrocyte differentiation (Domingues et al., 2020). These results largely indicate that MYO1E is strongly implicated in various processes associated with myelination.

Study Limitations: A limitation of this study is that it did not target regions of interest specifically parcellated for white matter tracts. Future research using white matter-specific parcellation methods is recommended to increase anatomical precision and better capture microstructural patterns contributing to ASD pathology.

Methods

Data Acquisition and Preprocessing: The study included 148 boys and girls with ASD (70 female, mean age = 150.8 months), and 175 neurotypical controls (98 female, mean age = 152.3 months), totaling 323 participants. Diffusion-weighted, T1, and T2-weighted images were collected from each participant as part of the NIH Autism Centers for Excellence Wave 1 cohort. Diffusion scans had a voxel size of 2x2x2mm with 64 gradient directions. T1 images had a field of view of 176x256x256mm, voxel size of 1x1x1mm, TE=3.3ms; T2 images had a field of view of 128x128x34mm, voxel size of 1.5x1.5x4mm, TE=35ms. Images were processed using the MICA-MNI pipeline, and voxel-wise T1/T2 ratios were calculated.

Generating White Matter Microstructural Similarity Networks (wm-MS): Based off of the morphometric similarity network, (Seidlitz et al., 2018) a new type of similarity network has been designed - the white matter microstructural similarity network (wm-MS). This metric will identify which regions of the brain - cortical, intra-cortical, and subcortical - contain distinct or similar features of white matter microstructure. A feature matrix consisting of three white matter microstructural features—conduction velocity, g-ratio, and white matter volume (Newman et al., 2024) —was estimated for all 323 subjects across 165 brain regions. Each feature vector in each region was Z-scored, and the Pearson correlation coefficient (r) was calculated for each regional pair.

Identifying Case-Control Differences in White Matter Microstructure: The regional mean for the *i*th participant at each region is the average of the *i*th row (or column) of the white matter microstructural similarity matrix (wm-MS). Regional wm-MS strength reflects the weighted degree, or hubness, of each node, connected by signed and weighted edges representing pairwise similarity across the brain. Linear models were used to estimate case-control differences in average regional wm-MS, adjusting for age, sex, and diagnosis.

Visualizing White Matter Microstructural Similarity Networks (wm-MS): New software, **Connectographics**, has designed and visualized Circos-based white matter microstructural similarity networks (wm-MS) between boys and girls with ASD and controls. Wm-MS Pearson correlation coefficients were thus re-converted to Z-scores to efficiently identify a wider range of stronger or weaker connections in the connectogram.

Identifying Genes Driving Case-Control Differences: Genes influencing case-control white matter differences were identified using the AHBA microarray dataset (Shen et al., 2012) to various regions of the human brain. Abagen software (Markello et al., 2021) was used to identify regional expression levels for 15,329 genes in Destrieux atlas parcellations. A subsequent partial least-squares regression analysis (PLS) was implemented (Morgan et al., 2019; Vertes et al., 2016) to relate case-control white matter differences in 165 brain regions, reflecting both subcortical regions and regions of the Destrieux atlas to microarray gene expression data. PLS uses microarray gene expression (predictor variables) to predict regional white matter microstructural similarity case-control t-statistics (response variables). PLS significance was tested via 1,000 permutations, and gene contributions were ranked by Z-scores based on bootstrap standard errors.

Figure 2: Visualizing Significant Brain Regions Linked to Case-Control Differences

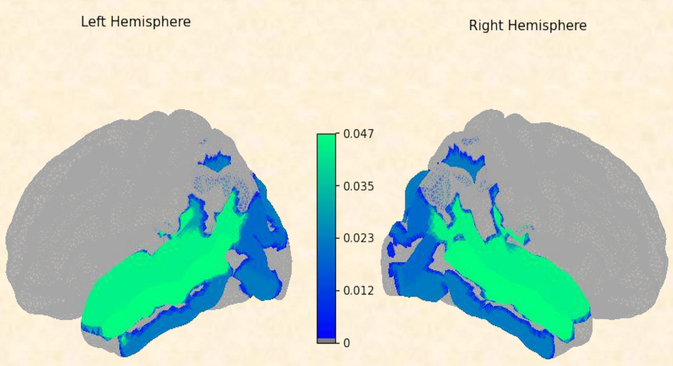


Figure 2 is a surface plot visualizing brain regions that exhibit significant differences in children with Autism compared to neurotypical children with respect to various microstructural features, specifically features such as g-ratio, conduction velocity, and white matter-related characteristics such as white-matter volume. Regions that confer substantial significance are in blue ($p < 0.02$) and regions of marginal significance are in light green ($0.05 > p > 0.04$).

Figure 3: Visualizing Significant Genes Linked to Case-Control Differences

Figure 3 visualizes the chromosomal locations of genes identified as significantly weighted in either PLS1 or PLS2 components (Z-score > 1.96) through Partial Least Squares (PLS) analysis. Genes are positioned along their respective chromosomes. Autism-related genes from the SFARI (Simons Foundation Autism Research Initiative) gene list are identified with a star symbol (*) distinguishing them as genes with known associations to ASD. The grey regions listed below are densities of structural variants (insertions, deletions, translocations) that have been extracted from the New York Genome Institute.

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